

2026 SISBID PCA Demo

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Setup

```
knitr::opts_chunk$set(  
  echo = TRUE,  
  message = FALSE,  
  warning = FALSE,  
  fig.align = "center",  
  fig.width = 7,  
  fig.height = 5  
)  
  
needed_packages <- c("ggplot2", "GGally")  
missing_packages <- needed_packages[!vapply(needed_packages, requireNamespace, logical(1), quietly = TRUE)]  
if (length(missing_packages) > 0) {  
  stop("Please install these packages before knitting: ", paste(missing_packages, collapse = ", "))  
}  
  
library(ggplot2)  
library(GGally)  
  
theme_set(theme_minimal(base_size = 12))  
set.seed(1)
```

Helper functions used throughout the demo:

```
variance_table <- function(pca_object) {  
  variance <- pca_object$sdev^2  
  pct <- 100 * variance / sum(variance)  
  data.frame(  
    component = seq_along(pct),  
    pct_variance = pct,  
    cumulative_pct_variance = cumsum(pct)  
  )  
}  
  
plot_variance <- function(var_tbl, title = "Variance explained", max_components = 30) {  
  keep <- seq_len(min(max_components, nrow(var_tbl)))  
  plot_df <- rbind(  
    data.frame(  
      component = var_tbl$component[keep],  
      value = var_tbl$pct_variance[keep],  
      metric = "Variance explained"  
    ),  
    data.frame(  

```

```

    component = var_tbl$component[keep],
    value = var_tbl$cumulative_pct_variance[keep],
    metric = "Cumulative variance explained"
  )
)

ggplot(plot_df, aes(x = component, y = value)) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 1.8) +
  facet_wrap(~ metric, ncol = 1, scales = "free_y") +
  labs(
    title = title,
    x = "Principal component",
    y = "Percent"
  )
}

top_loadings <- function(pca_object, component = 1, n = 8) {
  loadings <- pca_object$rotation[, component]
  ord <- order(abs(loadings), decreasing = TRUE)
  ord <- ord[seq_len(min(n, length(ord)))]
  data.frame(
    component = paste0("PC", component),
    variable = names(loadings)[ord],
    loading = round(loadings[ord], 3),
    absolute_loading = round(abs(loadings[ord]), 3),
    row.names = NULL
  )
}

```

PCA Demo Using Digits Data

Load digits data

```

data_file <- "UnsupL_SISBID_2026.Rdata"
if (!file.exists(data_file)) {
  stop("Expected data file not found: ", data_file)
}

load(data_file)

if (!exists("digits")) {
  stop("The data file must contain an object named 'digits'.")
}

if (!exists("imagedigit")) {
  imagedigit <- function(x) {
    side <- sqrt(length(x))
    if (side != floor(side)) {
      stop("Cannot display image: digit vector length is not a perfect square.")
    }
    mat <- matrix(x, nrow = side, byrow = TRUE)
  }
}

```

```

    image(t(apply(mat, 2, rev)), axes = FALSE, col = gray.colors(256))
  }
}

digits <- as.matrix(digits)
digit_labels <- rownames(digits)

if (is.null(digit_labels)) {
  stop("The digits matrix must have row names containing digit labels.")
}

digit_keep <- digit_labels %in% c("3", "8")
dat38 <- digits[digit_keep, , drop = FALSE]
dat38_labels <- factor(digit_labels[digit_keep], levels = c("3", "8"))

cat("Number of selected digit images:", nrow(dat38), "\n")

## Number of selected digit images: 1532
cat("Number of pixels per image:", ncol(dat38), "\n")

## Number of pixels per image: 256
table(dat38_labels)

## dat38_labels
##    3    8
## 824 708

```

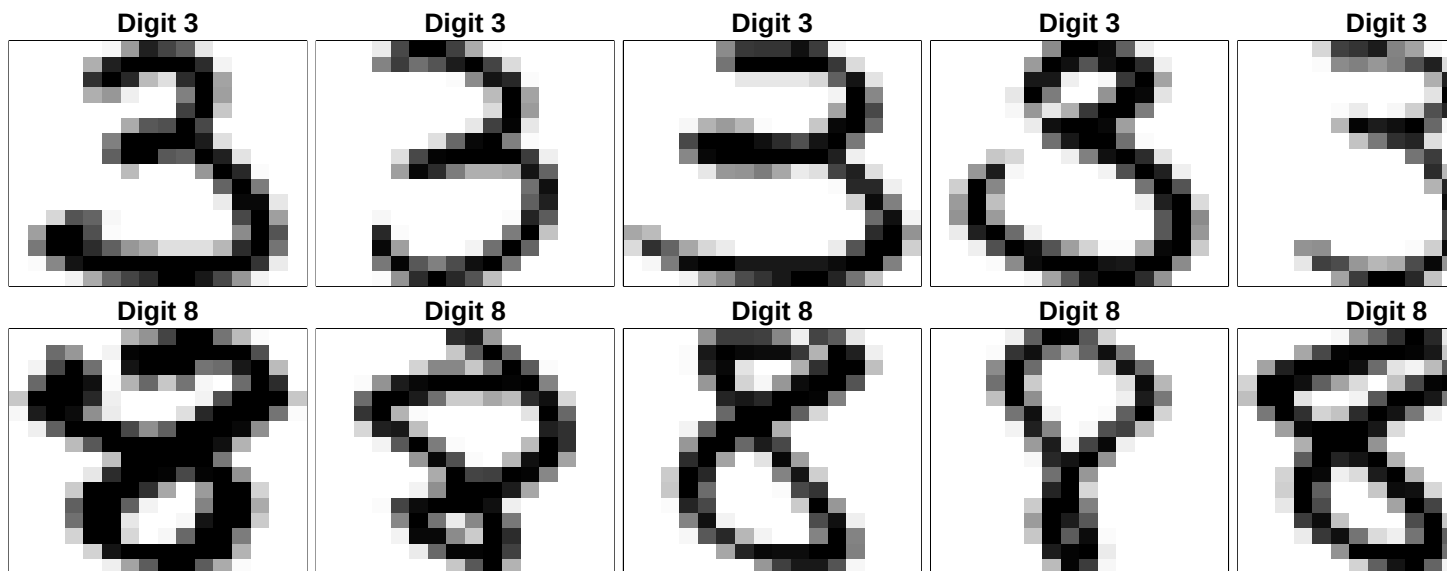
Visualize a few original images

```

example_ids <- c(which(dat38_labels == "3")[1:5], which(dat38_labels == "8")[1:5])
example_ids <- example_ids[!is.na(example_ids)]

old_par <- par(mfrow = c(2, 5), mar = c(0.2, 0.2, 1.5, 0.2))
for (i in example_ids) {
  imagedigit(dat38[i, ])
  title(main = paste("Digit", dat38_labels[i]))
}

```



```
par(old_par)
```

PCA on centered digit pixels

For image data, we center each pixel before PCA. We do not scale pixels here because scaling can over-emphasize nearly constant background pixels. For data sets with variables on very different scales, see the College example below.

```
digit_pca <- prcomp(dat38, center = TRUE, scale. = FALSE)
digit_var <- variance_table(digit_pca)

head(digit_var, 10)
```

##	component	pct_variance	cumulative_pct_variance
## 1	1	15.158812	15.15881
## 2	2	8.731200	23.89001
## 3	3	6.674469	30.56448
## 4	4	5.154120	35.71860
## 5	5	4.967040	40.68564
## 6	6	3.984085	44.66973
## 7	7	3.447350	48.11708
## 8	8	2.892137	51.00921
## 9	9	2.638936	53.64815
## 10	10	2.295966	55.94412

PC scatterplot

```
digit_score_mat <- digit_pca$x[, 1:5, drop = FALSE]
rownames(digit_score_mat) <- NULL
digit_scores <- as.data.frame(digit_score_mat)
digit_scores$digit <- dat38_labels

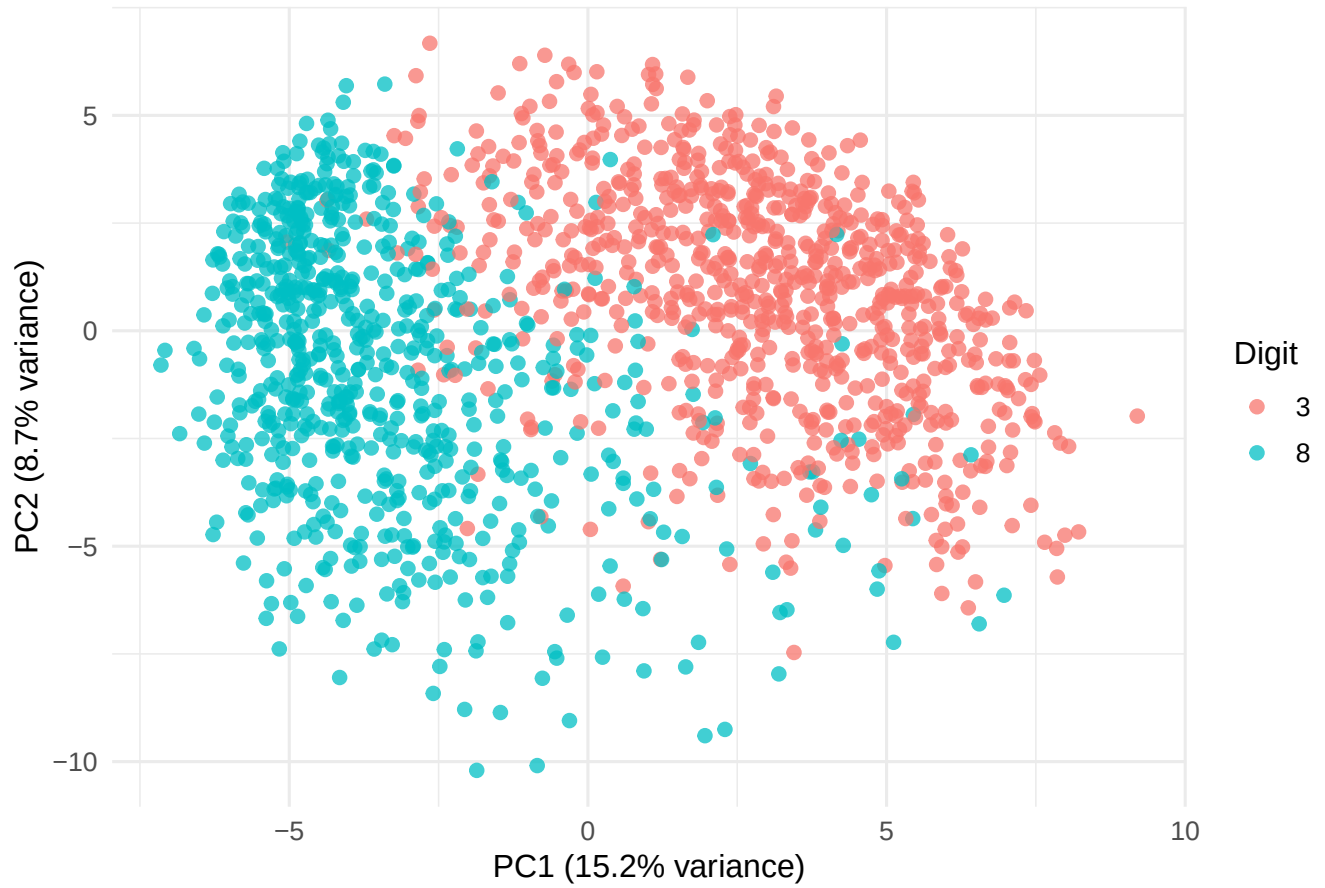
ggplot(digit_scores, aes(x = PC1, y = PC2, color = digit)) +
  geom_point(alpha = 0.75, size = 2) +
  labs(
    title = "Digits 3 and 8: first two principal components",
```

```

x = paste0("PC1 (", round(digit_var$pct_variance[1], 1), "% variance)",
y = paste0("PC2 (", round(digit_var$pct_variance[2], 1), "% variance)",
color = "Digit"
)

```

Digits 3 and 8: first two principal components



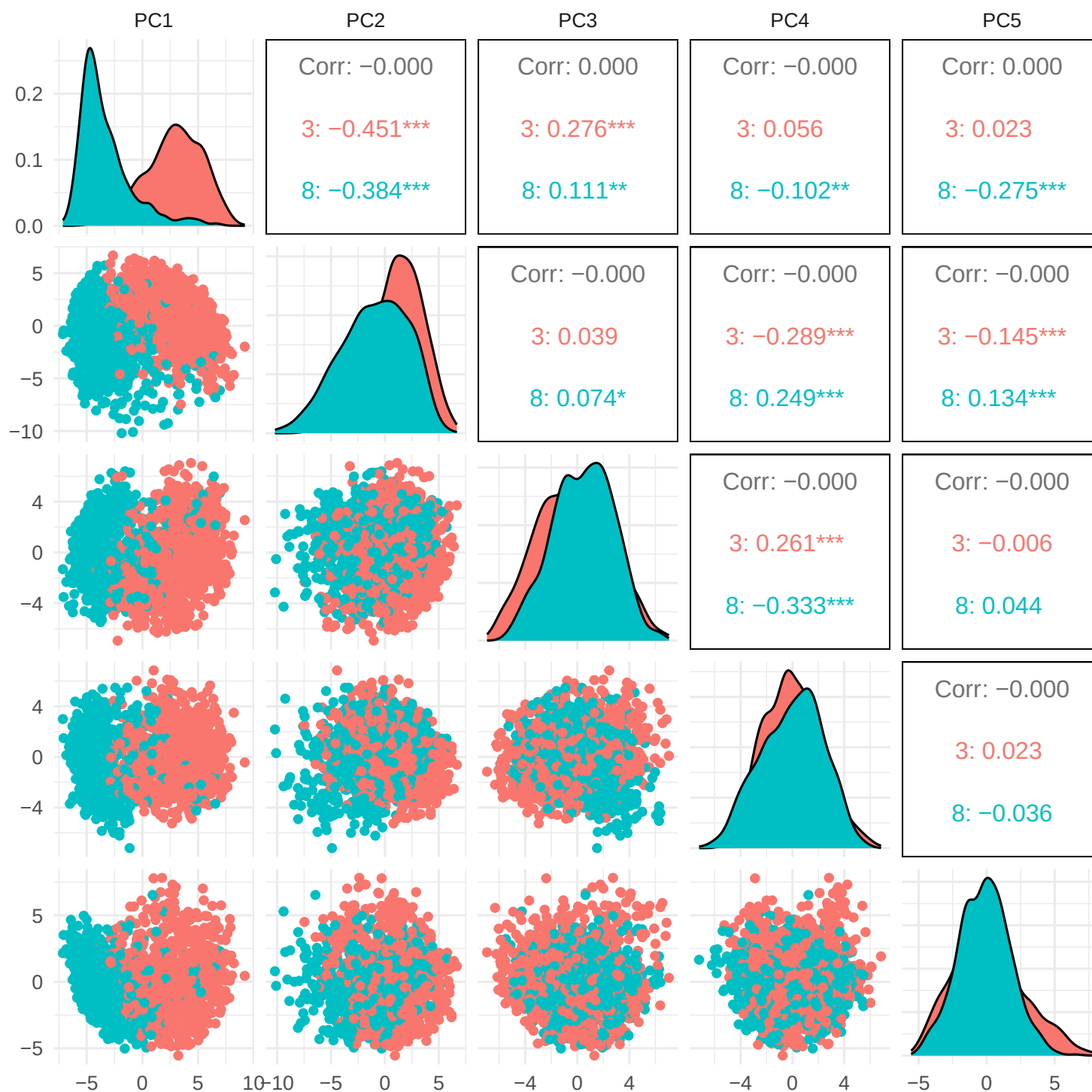
Pairs plot for the first five PCs

```

ggpairs(
  digit_scores,
  columns = 1:5,
  mapping = aes(color = digit),
  title = "Pairwise plots of the first five digit PCs"
)

```

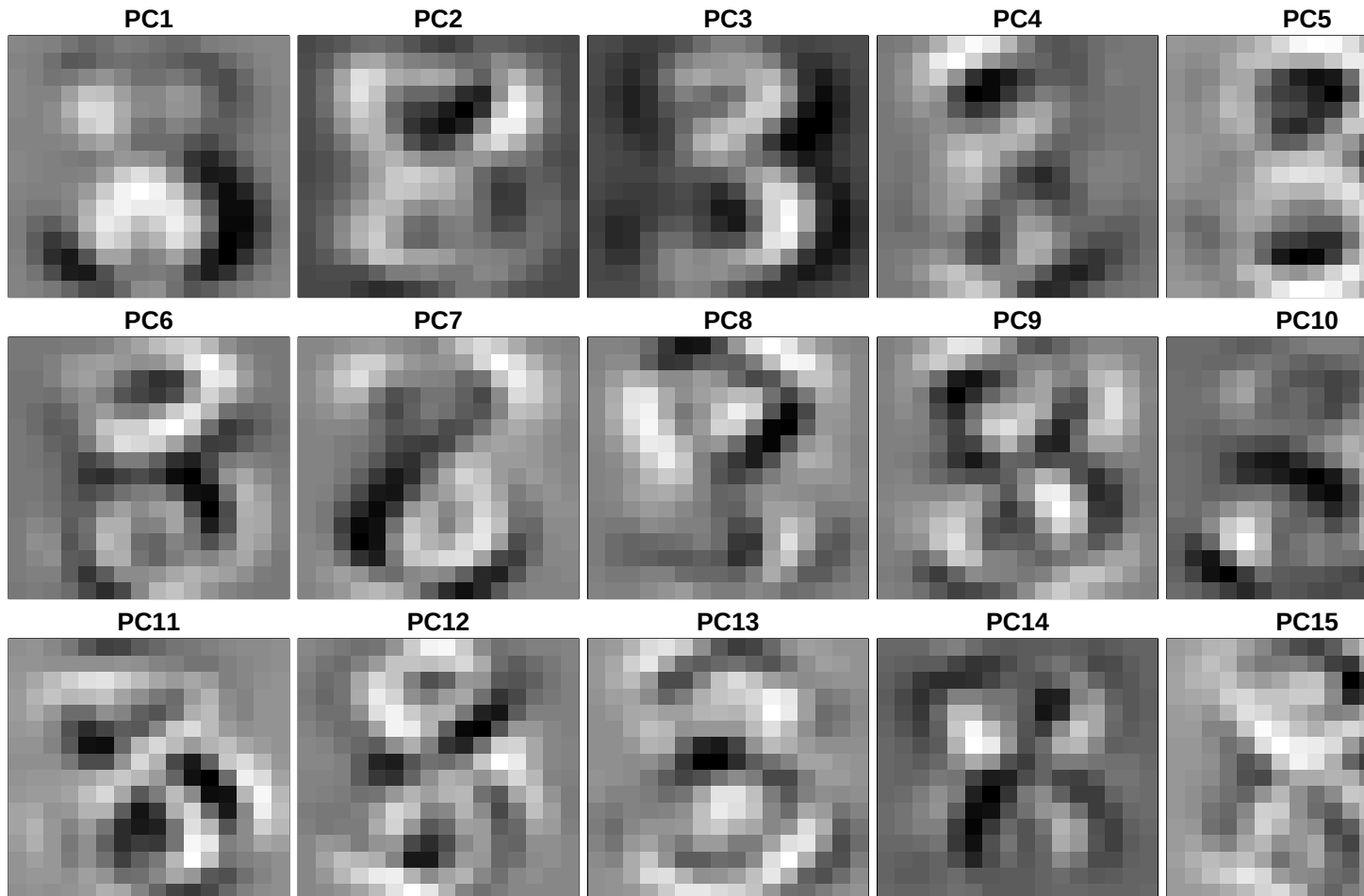
Pairwise plots of the first five digit PCs



PC Loadings

Each loading image shows a direction of variation in pixel space. Bright and dark regions indicate pixels that move in opposite directions along that component.

```
old_par <- par(mfrow = c(3, 5), mar = c(0.2, 0.2, 1.5, 0.2))
for (i in 1:15) {
  imagedigit(digit_pca$rotation[, i])
  title(main = paste0("PC", i))
}
```

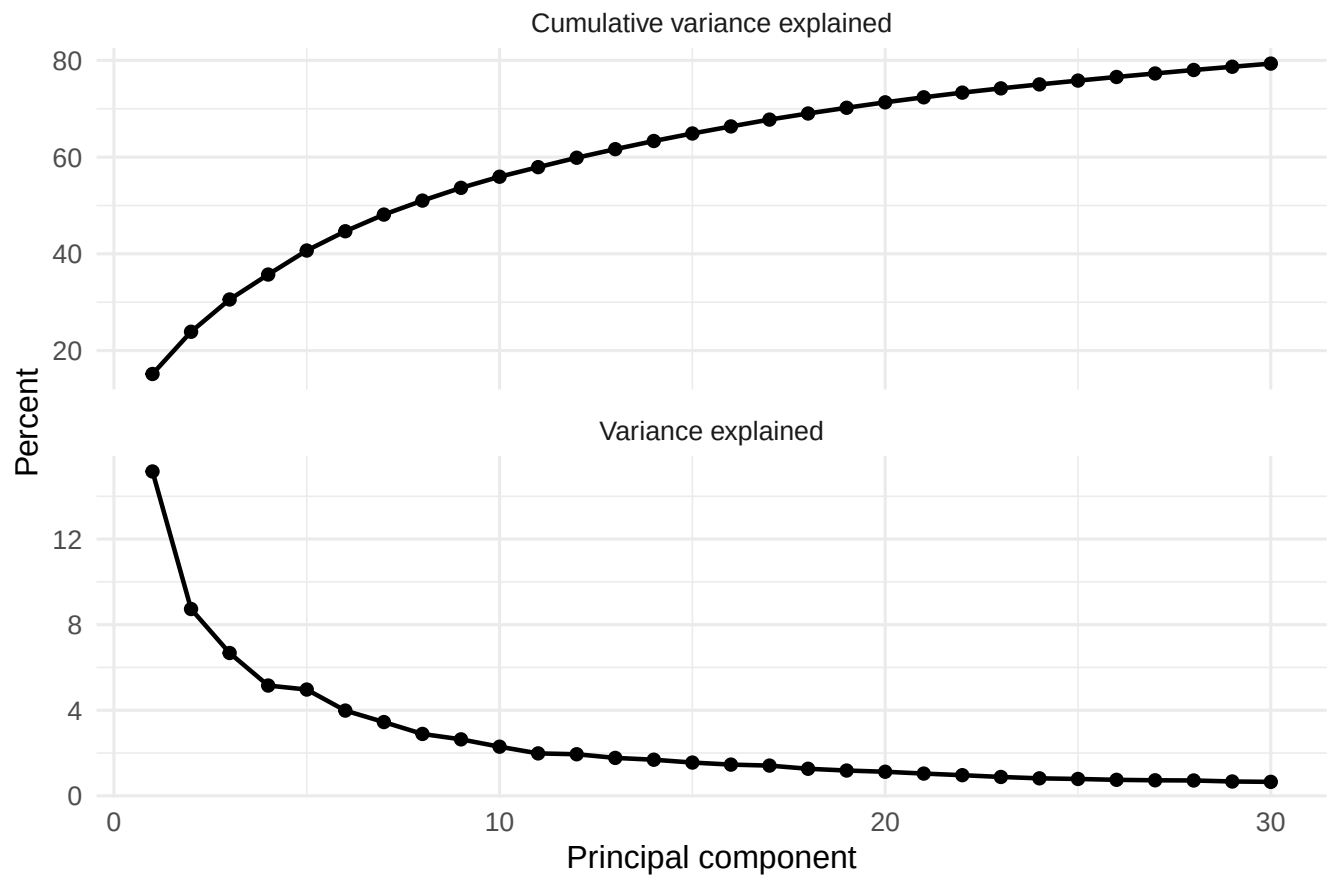


```
par(old_par)
```

Variance explained

```
plot_variance(digit_var, title = "Digit PCA variance explained", max_components = 30)
```

Digit PCA variance explained



```
thresholds <- c(50, 75, 90, 95)
components_needed <- sapply(thresholds, function(thresh) {
  which(digit_var$cumulative_pct_variance >= thresh)[1]
})

data.frame(
  cumulative_variance_threshold = paste0(thresholds, "%"),
  components_needed = components_needed
)
```

```
## cumulative_variance_threshold components_needed
## 1 50% 8
## 2 75% 24
## 3 90% 58
## 4 95% 89
```

Low-rank reconstruction

```
reconstruct_digit <- function(row_id, n_components) {
  scores <- digit_pca$x[row_id, 1:n_components, drop = FALSE]
  loadings <- digit_pca$rotation[, 1:n_components, drop = FALSE]
  reconstruction <- scores %*% t(loadings)
  as.numeric(reconstruction + digit_pca$center)
}
```



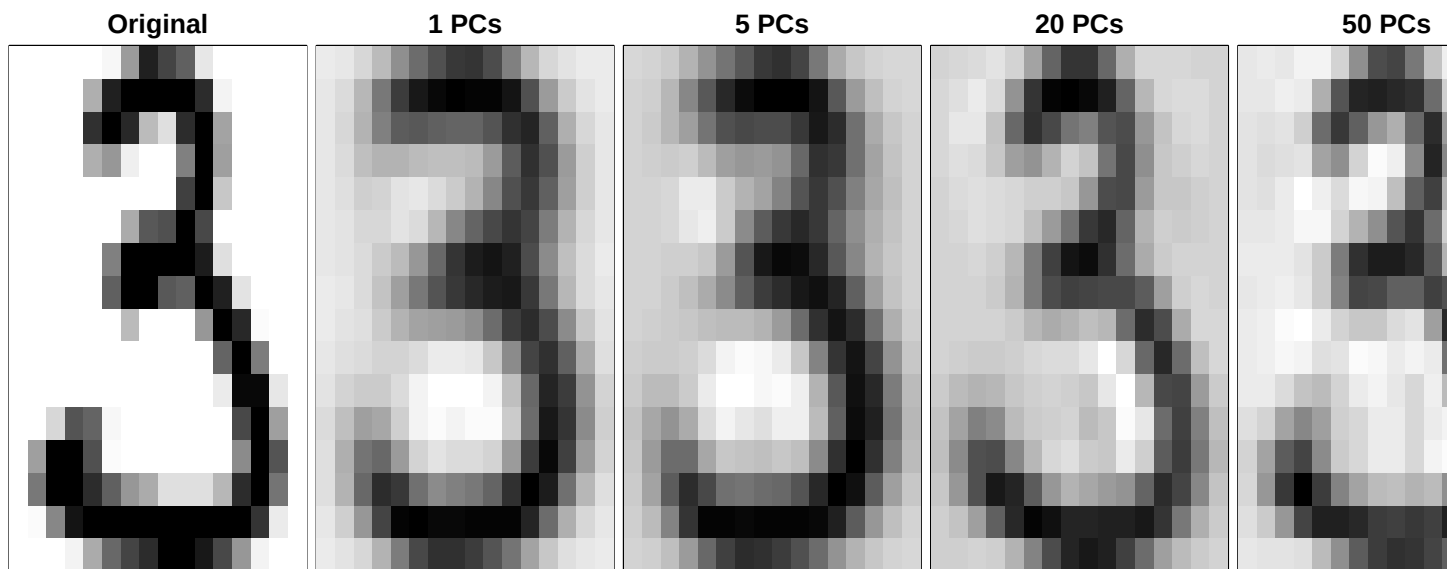
```

row_id <- which(dat38_labels == "3")[1]
component_grid <- c(1, 5, 20, 50)
component_grid <- component_grid[component_grid <= ncol(digit_pca$x)]

old_par <- par(mfrow = c(1, length(component_grid) + 1), mar = c(0.2, 0.2, 1.8, 0.2))
imagedigit(dat38[row_id, ])
title(main = "Original")

for (k in component_grid) {
  imagedigit(reconstruct_digit(row_id, k))
  title(main = paste(k, "PCs"))
}

```



```
par(old_par)
```

PCA via SVD

Raw SVD without centering

```

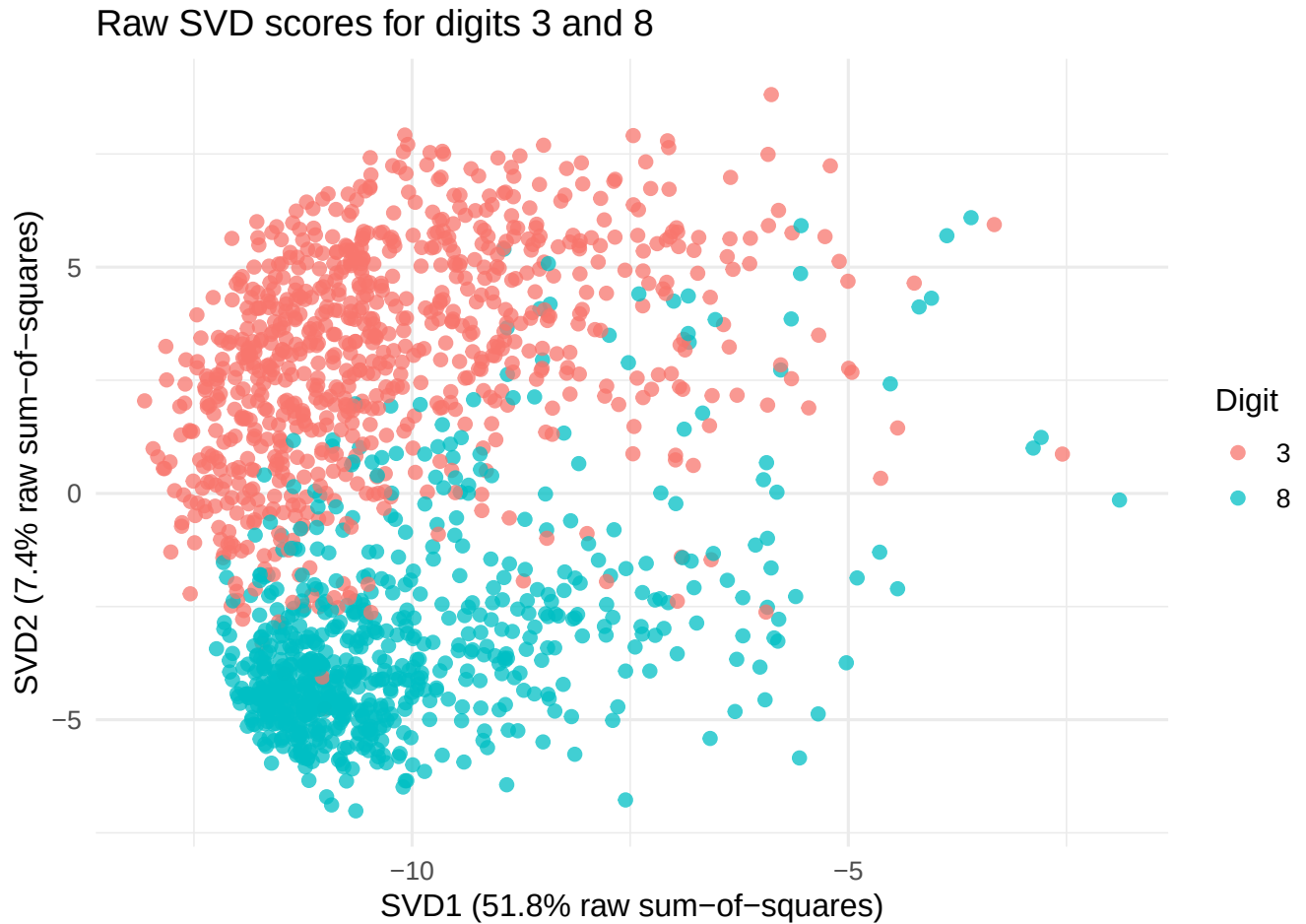
svd_raw <- svd(dat38)
raw_scores <- sweep(svd_raw$u[, 1:5, drop = FALSE], 2, svd_raw$d[1:5], `*`)
colnames(raw_scores) <- paste0("SVD", 1:5)
raw_score_df <- as.data.frame(raw_scores)
raw_score_df$digit <- dat38_labels

raw_var <- data.frame(
  component = seq_along(svd_raw$d),
  pct_variance = 100 * svd_raw$d^2 / sum(svd_raw$d^2)
)
raw_var$cumulative_pct_variance <- cumsum(raw_var$pct_variance)

ggplot(raw_score_df, aes(x = SVD1, y = SVD2, color = digit)) +
  geom_point(alpha = 0.75, size = 2) +

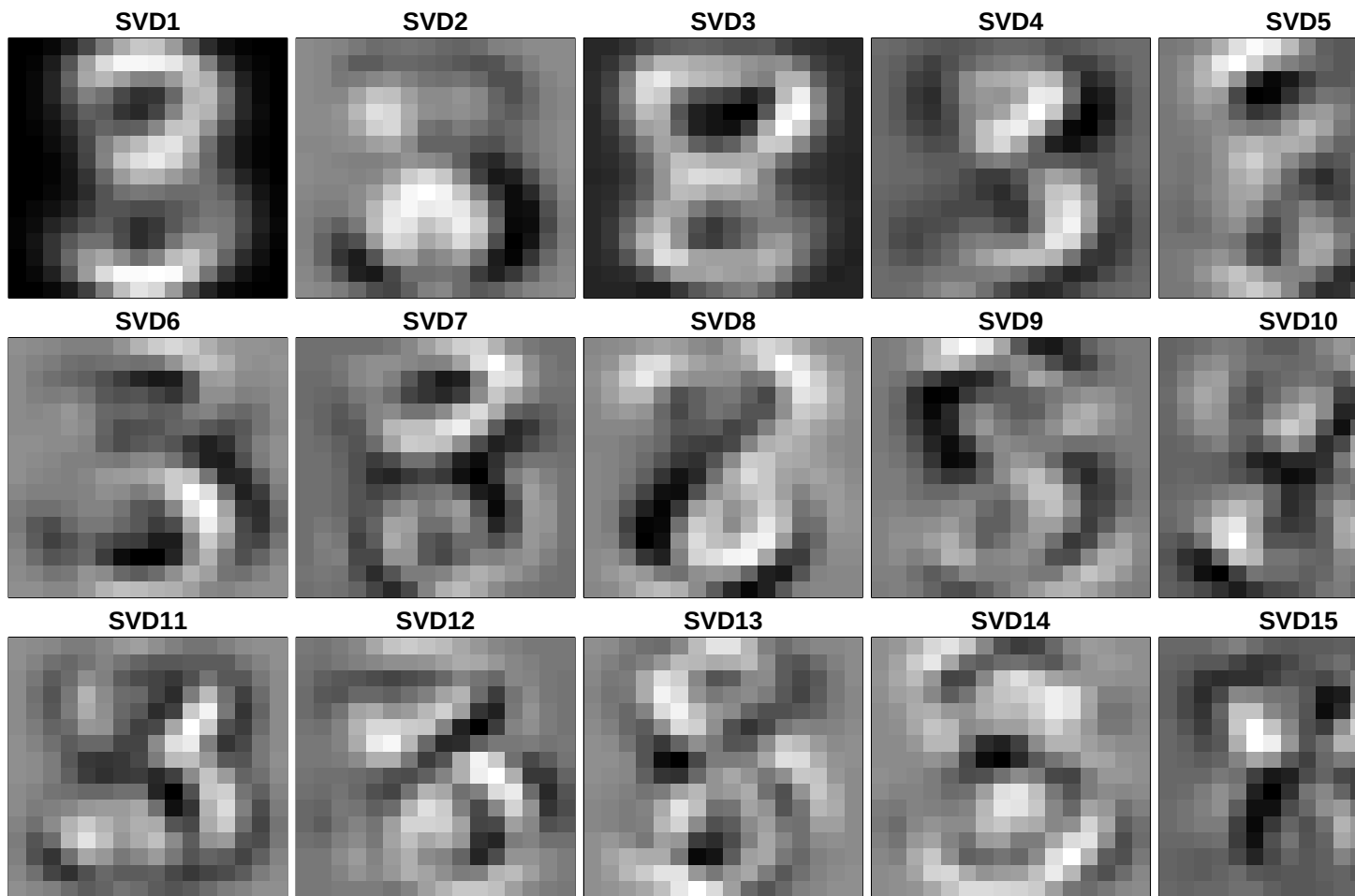
```

```
labs(
  title = "Raw SVD scores for digits 3 and 8",
  x = paste0("SVD1 (", round(raw_var$pct_variance[1], 1), "% raw sum-of-squares)"),
  y = paste0("SVD2 (", round(raw_var$pct_variance[2], 1), "% raw sum-of-squares)"),
  color = "Digit"
)
```



SVD Loadings

```
old_par <- par(mfrow = c(3, 5), mar = c(0.2, 0.2, 1.5, 0.2))
for (i in 1:15) {
  imagedigit(svd_raw$v[, i])
  title(main = paste0("SVD", i))
}
```

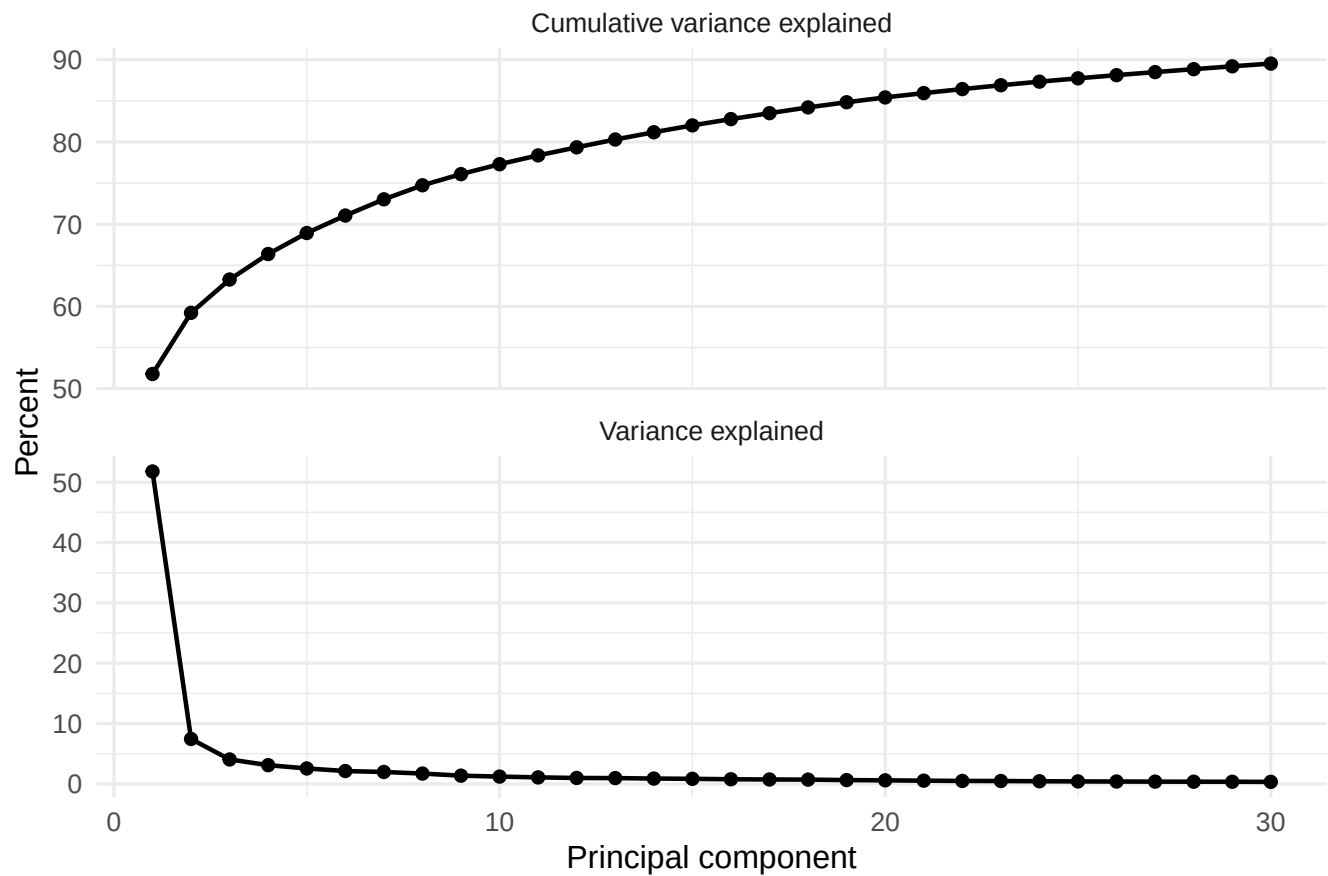


```
par(old_par)
```

Raw SVD variance explained

```
plot_variance(raw_var, title = "Raw SVD variance explained", max_components = 30)
```

Raw SVD variance explained



Centered SVD and PCA equivalence check

```
centered_digits <- scale(dat38, center = TRUE, scale = FALSE)
svd_centered <- svd(centered_digits)

# Singular vectors are identifiable only up to sign, so compare absolute values.
max_loading_difference <- max(abs(abs(svd_centered$v[, 1]) - abs(digit_pca$rotation[, 1])))
max_loading_difference
```

```
## [1] 0
```

PCA Demo Using College Data

Load College data

```
load_college_data <- function() {
  env <- new.env(parent = emptyenv())

  if (requireNamespace("ISLR", quietly = TRUE)) {
    data("College", package = "ISLR", envir = env)
    return(get("College", envir = env))
  }
}
```

```

if (requireNamespace("ISLR2", quietly = TRUE)) {
  data("College", package = "ISLR2", envir = env)
  return(get("College", envir = env))
}

stop("Please install either the ISLR or ISLR2 package to use the College data.")
}

College <- load_college_data()
college_numeric <- vapply(College, is.numeric, logical(1))
cdat <- College[, college_numeric, drop = FALSE]

cat("College data dimensions:", dim(cdat), "\n")

## College data dimensions: 777 17

names(cdat)

## [1] "Apps"          "Accept"        "Enroll"        "Top10perc"     "Top25perc"
## [6] "F.Undergrad"   "P.Undergrad"   "Outstate"      "Room.Board"    "Books"
## [11] "Personal"      "PhD"           "Terminal"      "S.F.Ratio"     "perc.alumni"
## [16] "Expend"        "Grad.Rate"

```

PCA with centering and scaling

For the College data, centering and scaling are important because variables such as enrollment counts, graduation rates, and dollar amounts are measured on different scales.

```

college_pca <- prcomp(cdat, center = TRUE, scale. = TRUE)
college_var <- variance_table(college_pca)

head(college_var, 10)

```

```

##      component pct_variance cumulative_pct_variance
## 1           1    32.020628             32.02063
## 2           2    26.340214             58.36084
## 3           3     6.900917             65.26176
## 4           4     5.922989             71.18475
## 5           5     5.488405             76.67315
## 6           6     4.984701             81.65785
## 7           7     3.558871             85.21673
## 8           8     3.453621             88.67035
## 9           9     3.117234             91.78758
## 10          10     2.375192             94.16277

```

PC scatterplot

```

college_scores <- as.data.frame(college_pca$x[, 1:5, drop = FALSE])
college_scores$college <- rownames(College)
college_scores$Private <- College$Private

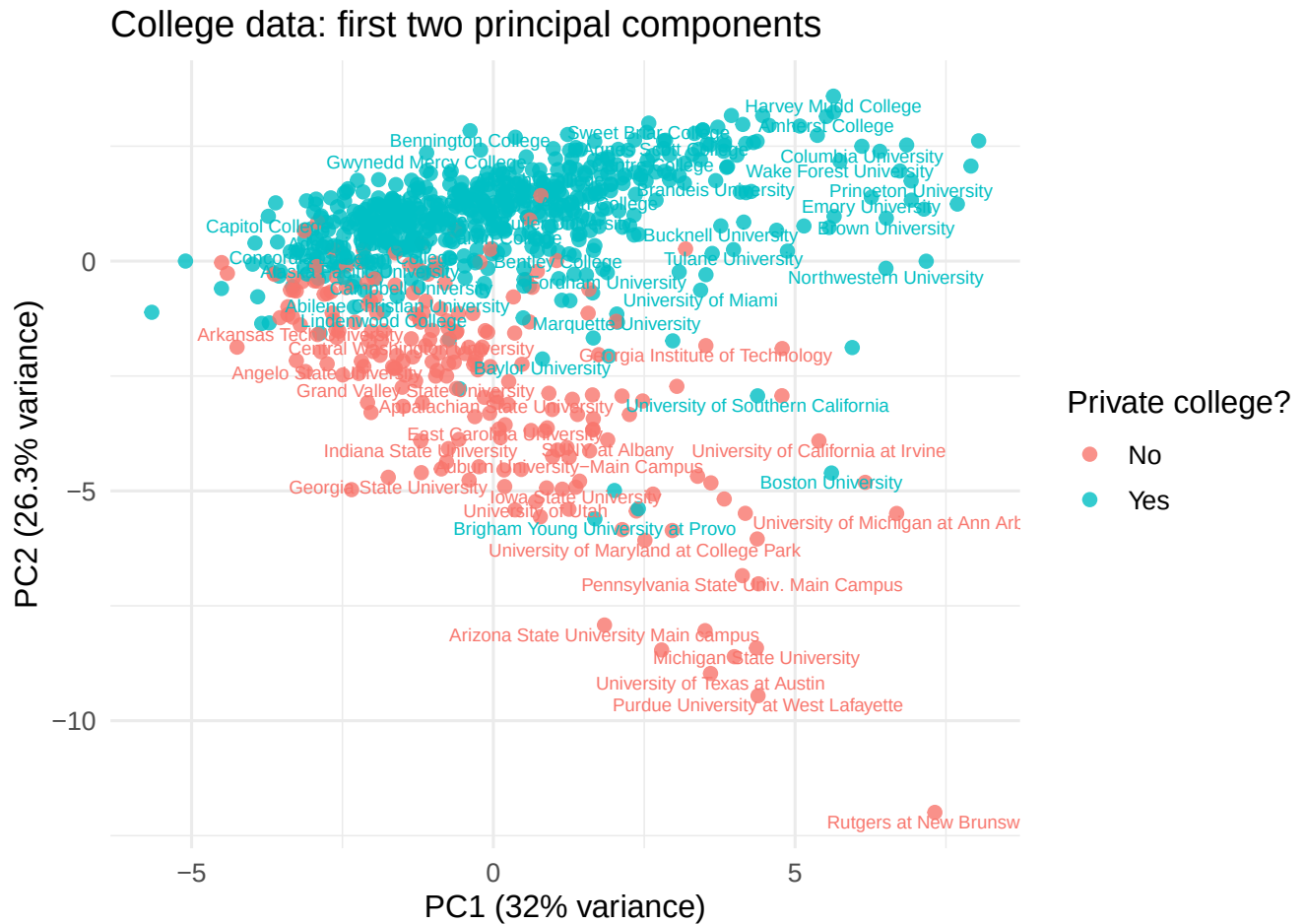
ggplot(college_scores, aes(x = PC1, y = PC2, color = Private, label = college)) +
  geom_point(alpha = 0.8, size = 2) +
  geom_text(check_overlap = TRUE, size = 2.5, vjust = 1.2, show.legend = FALSE) +
  labs(

```

```

title = "College data: first two principal components",
x = paste0("PC1 (", round(college_var$pct_variance[1], 1), "% variance)"),
y = paste0("PC2 (", round(college_var$pct_variance[2], 1), "% variance)"),
color = "Private college?"
)

```



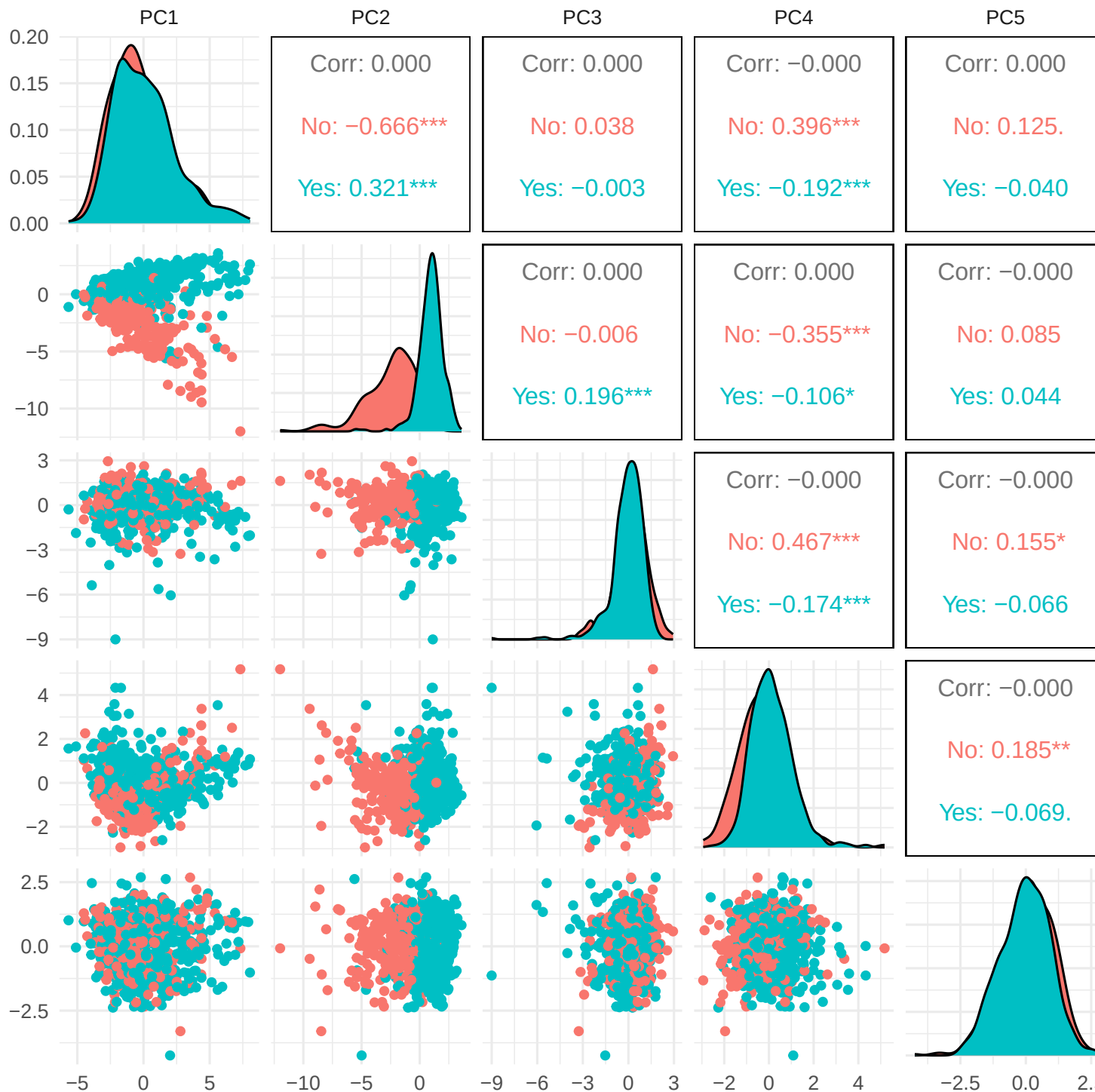
Pairs plot

```

ggpairs(
  college_scores,
  columns = 1:5,
  mapping = aes(color = Private),
  title = "Pairwise plots of the first five College PCs"
)

```

Pairwise plots of the first five College PCs



Loadings: variables that contribute to the patterns

```
college_top_loadings <- rbind(
  top_loadings(college_pca, component = 1, n = 8),
  top_loadings(college_pca, component = 2, n = 8)
```

```
)
```

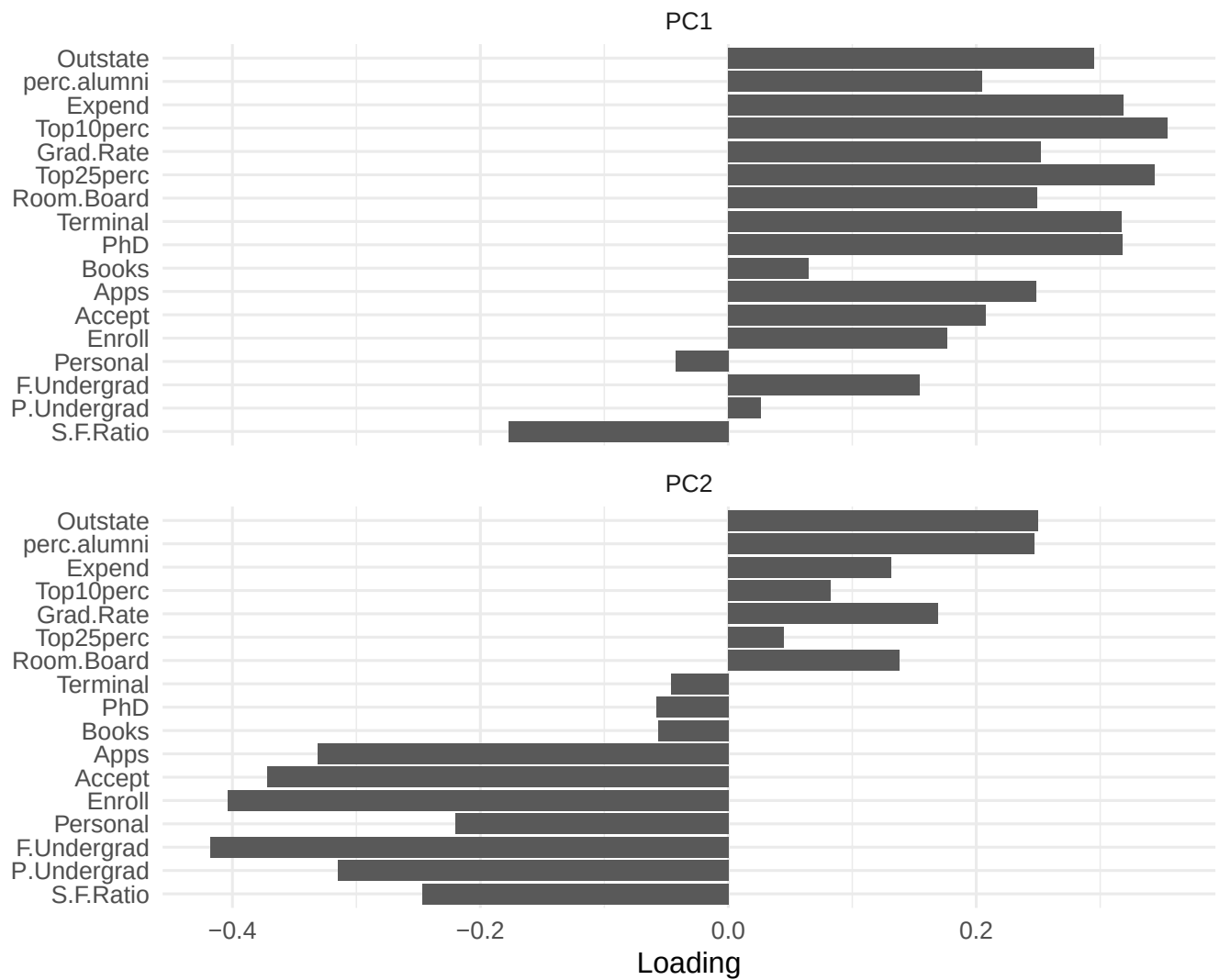
```
knitr::kable(college_top_loadings)
```

component	variable	loading	absolute_loading
PC1	Top10perc	0.354	0.354
PC1	Top25perc	0.344	0.344
PC1	Expend	0.319	0.319
PC1	PhD	0.318	0.318
PC1	Terminal	0.317	0.317
PC1	Outstate	0.295	0.295
PC1	Grad.Rate	0.252	0.252
PC1	Room.Board	0.249	0.249
PC2	F.Undergrad	-0.418	0.418
PC2	Enroll	-0.404	0.404
PC2	Accept	-0.372	0.372
PC2	Apps	-0.332	0.332
PC2	P.Undergrad	-0.315	0.315
PC2	Outstate	0.250	0.250
PC2	S.F.Ratio	-0.247	0.247
PC2	perc.alumni	0.247	0.247

```
loading_df <- rbind(
  data.frame(
    variable = rownames(college_pca$rotation),
    loading = college_pca$rotation[, 1],
    component = "PC1"
  ),
  data.frame(
    variable = rownames(college_pca$rotation),
    loading = college_pca$rotation[, 2],
    component = "PC2"
  )
)

ggplot(loading_df, aes(x = reorder(variable, loading), y = loading)) +
  geom_col() +
  coord_flip() +
  facet_wrap(~ component, ncol = 1, scales = "free_y") +
  labs(
    title = "College PCA loadings for PC1 and PC2",
    x = NULL,
    y = "Loading"
  )
```

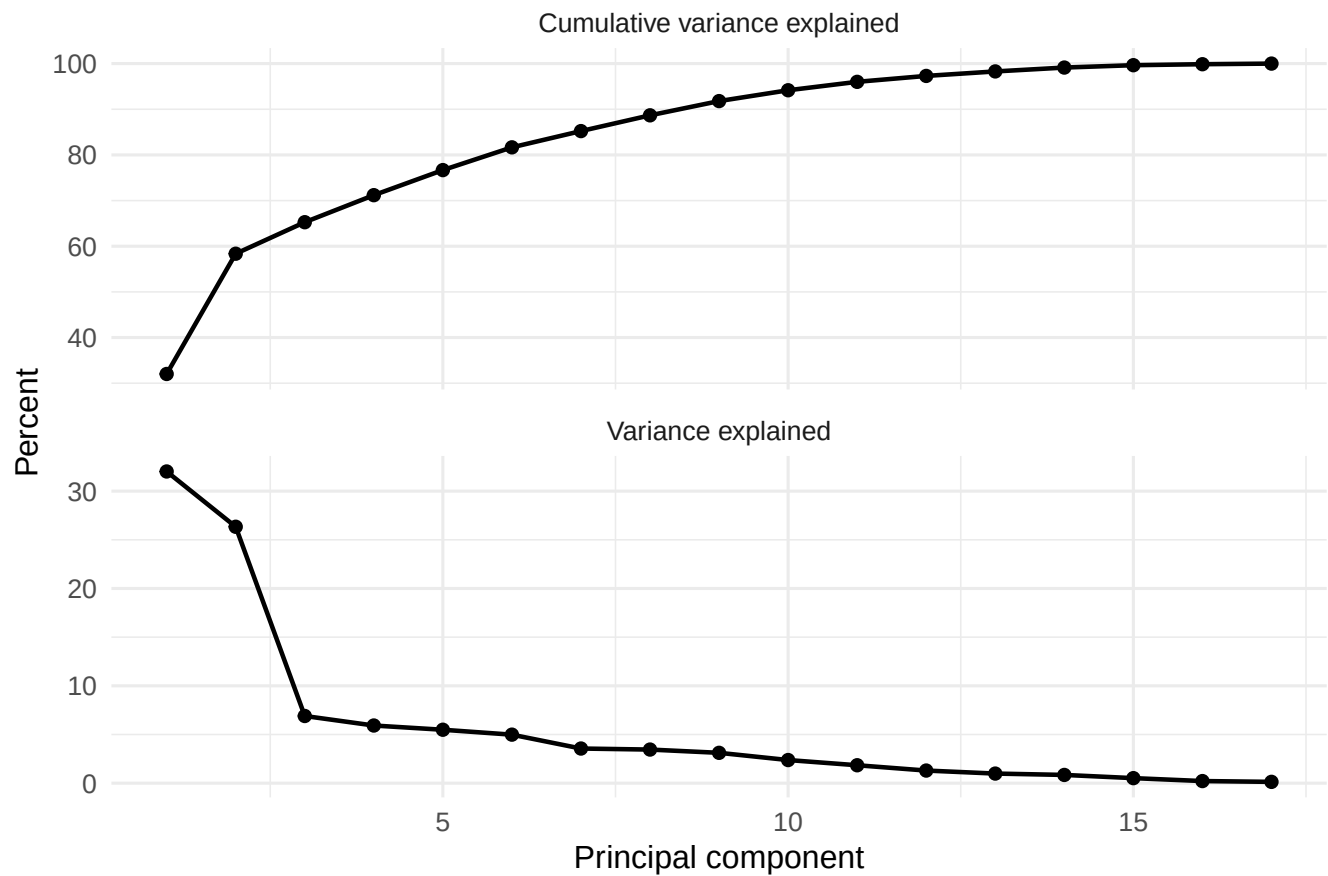

College PCA loadings for PC1 and PC2



Variance explained

```
plot_variance(college_var, title = "College PCA variance explained", max_components = nrow(college_var))
```

College PCA variance explained



```
thresholds <- c(50, 75, 90, 95)
components_needed <- sapply(thresholds, function(thresh) {
  which(college_var$cumulative_pct_variance >= thresh)[1]
})

data.frame(
  cumulative_variance_threshold = paste0(thresholds, "%"),
  components_needed = components_needed
)
```

```
## cumulative_variance_threshold components_needed
## 1 50% 2
## 2 75% 5
## 3 90% 9
## 4 95% 11
```